

Award Accounts

The Chemical Society of Japan Award for Young Chemists for 2014

Development of Neighboring Electrophilic Activation of Active Center in Catalytic Reactions via Organometallic Intermediates

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Received: January 12, 2017; Accepted: February 16, 2017; Web Released: April 25, 2017



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Kohei Endo received his Ph.D. from The University of Tokyo in 2006. He was a postdoctoral fellow at Nagoya University (April 2006 through March 2007), and then moved to Waseda University as an assistant professor in April 2007. From April 2011, he was a PRESTO researcher at the Japan Science and Technology Agency (JST) until March 2016. In January 2012, he moved to Kanazawa University as Associate Professor. He has now been Associate Professor at Tokyo University of Science since April 2015.

Abstract

Strategies for electrophilic activation by neighboring atoms should contribute to the development of novel catalytic performance in organic reactions. Neighboring electron-withdrawing Lewis acidic atoms can change the character of molecular catalysts, which might be more electrophilic toward inactive electron-rich molecules. In this context, the development of carbene intermediates, diborylmethane derivatives, and multi-nuclear catalysts has realized efficient transformations. The use of cyclopropanes can generate electrophilic carbenoid intermediates in the presence of a Ag-catalyst, the reaction of which with an organozinc reagent gives allylzinc intermediates. The subsequent allylation reaction of other electrophiles takes place in situ as a 3-components coupling reaction. The novel reactivity of diborylmethane derivatives shows the generation of borate intermediates under ambient conditions, which can take part in the Suzuki–Miyaura cross-coupling reaction. The mono-coupling reaction occurs exclusively without di-coupling reaction to give alkylboronates. The stereoselective synthesis of tetrasubstituted alkenylboronates was successful via the deprotonation of diborylalkanes, subsequent nucleophilic attack to ketones, and *syn*-elimination. The development of multinuclear catalysts using phosphorous ligands bearing protic moieties has realized efficient stereoselective 1,4-addition reaction of organozinc and organoaluminum reagents, where the creation of chiral quaternary stereogenic centers using acyclic enones is notable.

1. Introduction

Cooperative effects are attractive features for the development of novel molecular functions in organic chemistry.¹ The unique reactivity and/or selectivity that can be achieved with cooperative effects may realize efficient and/or unexpected transformations. Numerous studies using enzymes, including artificial enzymes, have shown cooperative effects for the activation of substrates in specific transformations.² We focused on molecules bearing Lewis acid moieties, which might change the electrostatic potentials as well as the orbital distributions in reactive intermediates (Figure 1). A reactive intermediate with electron-withdrawing Lewis acidic moieties may enhance the electrophilic character of an active center toward a nucleophilic attack. There have been reports on σ -acceptor ligands, i.e., Z-type ligands, where a vacant orbital arising from a Lewis acidic atom induces back-donation from a neighboring transition-metal center.³ Z-type ligands might provide higher catalytic activity, although the mechanism is unclear. Here we have

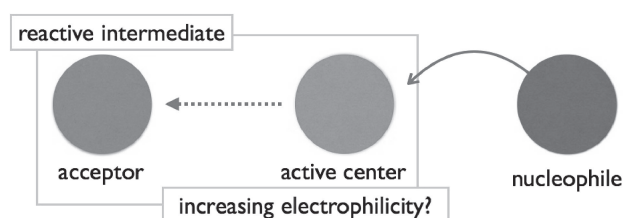


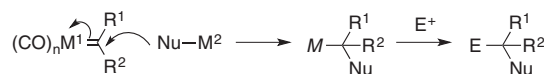
Figure 1. Electrophilic activation strategy.

focused on the design of electrophilic reactive intermediates in catalytic reactions with Lewis acidic moieties.

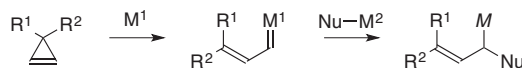
2. Results and Discussion

2.1 Development of Electrophilic Carbene Intermediates under Catalytic Conditions. The initial target of the present study is carbenoid intermediates, namely Fischer carbene-type intermediates, which are electrophilic organometallic intermediates bearing an electron-withdrawing metal center (Scheme 1).⁴ Typical Fischer-type carbene intermediates are carbene complexes bearing CO ligands, as a strong electron-withdrawing ligand, where the nucleophilic addition of organometallic reagents to a formal C–metal double bond takes place.⁵ The subsequent reaction with electrophiles achieves tandem 3-component coupling reactions.

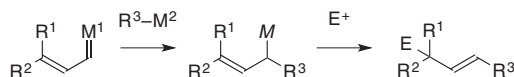
The generation of nucleophilic intermediates from Fischer-type carbene complexes in the presence of a catalyst and organometallic reagents would realize a tandem reaction with internal electrophiles. There have been several reports on the stepwise nucleophilic addition reaction via carbene intermediates.



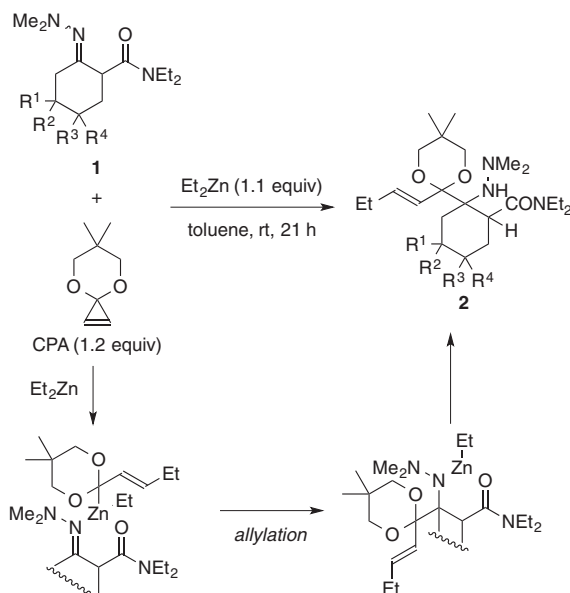
Scheme 1. Addition of organometallic intermediates to Fischer-type carbene complexes.



Scheme 2. Generation of carbene complexes and allylmetal intermediate derived from cyclopropenes.



Scheme 3. Concept of tandem nucleophilic addition.



Scheme 4. Addition to cyclopropenes/ring-opening reaction.

ates.⁵ We hypothesized that stable cyclopropenes are promising precursors for the generation of carbene intermediates in the presence of a Rh, Au, or Ag complex as a catalyst (Scheme 2). The catalytic generation of carbene intermediates and a subsequent nucleophilic addition would give an allylic metal intermediate for the later allylation reaction of an electrophile (Scheme 3). During the course of our studies, we found that the reaction using hydrazone derivatives **1**, cyclopropenone-acetal (CPA), and organozinc reagents took place to give 3-component coupling products **2** without any transition-metal catalysts.⁶ The nucleophilic addition to cyclopropenes, carbometallation, and ring-opening reaction of cyclopropylmetal intermediates was reported using organoaluminum reagents. Thus, we initially wondered if the present reaction would occur via the carbocationization of cyclopropenes and subsequent ring-opening to give an allylzinc intermediate, which undergoes the allylation reaction of hydrazones (Scheme 4, Figure 2). Extension of the present approach realized a tandem allylation

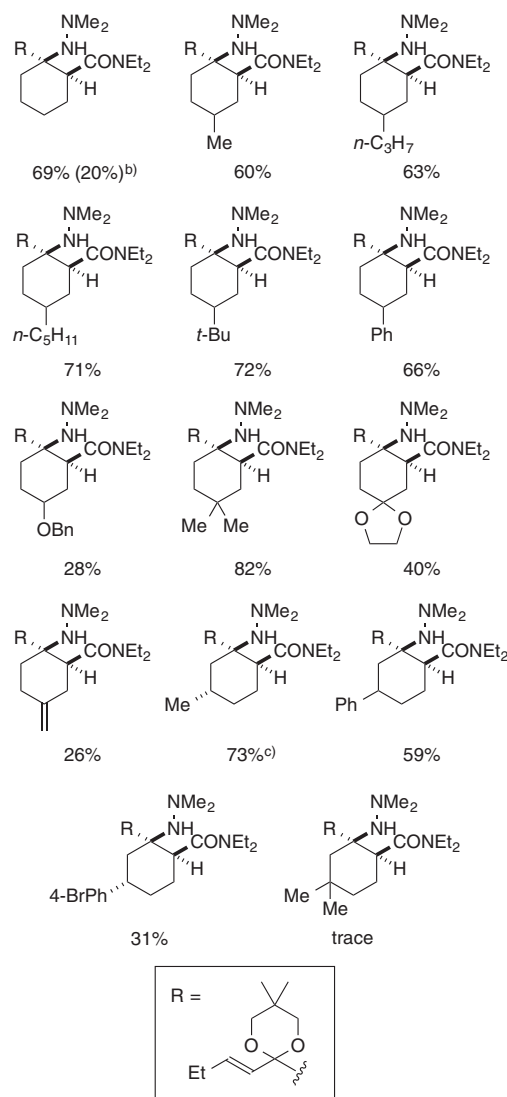
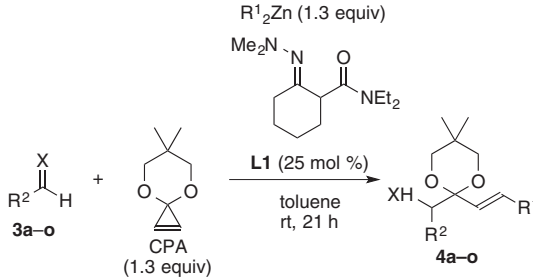


Figure 2. Products for tandem allylation^a. a) Single diastereomer was obtained, respectively. b) Me₂Zn was used instead of Et₂Zn. c) X-ray single crystallographic analysis was performed.

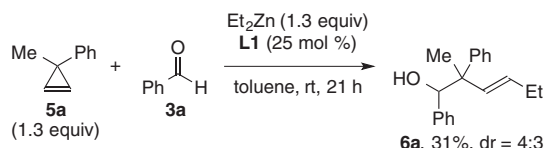
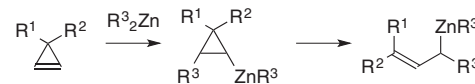
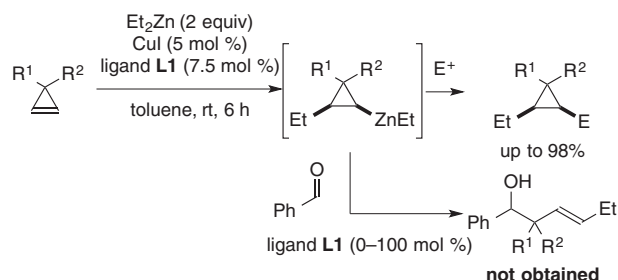
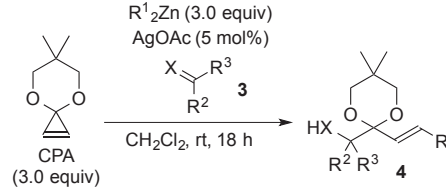
Table 1. Catalytic tandem allylation of aldehydes


Entry	R ¹ , R ² , X	Yield (%)
1	Et, Ph, O (3a)	76 (4a)
2	Me, Ph, O (3a)	48 (4a-Me)
3	<i>i</i> -Pr, Ph, O (3a)	63 (4a-i-Pr)
4	Et, 2-MeC ₆ H ₄ , O (3b)	62 (4b)
5	Et, 3-MeC ₆ H ₄ , O (3c)	68 (4c)
6	Et, 4-MeC ₆ H ₄ , O (3d)	71 (4d)
7	Et, 4-CF ₃ C ₆ H ₄ , O (3e)	70 (4e)
8	Et, 4-MeOC ₆ H ₄ , O (3f)	52 (4f)
9 ^a	Et, 4-FC ₆ H ₄ , O (3g)	56 (4g)
10	Et, 1-naphthyl, O (3h)	48 (4h)
11	Et, 2-naphthyl, O (3i)	trace (4i)
12	Et, 2-furyl, O (3j)	74 (4j)
13	Et, 2-phenylethyl, O (3k)	73 (4k)
14	Et, 2-phenylethenyl, O (3l)	68 (4l)
15	Et, <i>c</i> -Hex, O (3m)	77 (4m)
16	Et, <i>t</i> -Bu, O (3n)	35 (4n) ^b
17	Et, Ph, NTs (3o)	41 (4o)

a) **L1** (2 equiv), Et₂Zn (2 equiv), and CPA (2 equiv) were used.

b) The product was isolated as a ketone after deprotection of acetal moiety.

reaction of other electrophiles in the presence of the hydrazone catalyst **L1**. The reaction using CPA, organozinc reagents, and aldehydes in the presence of the hydrazone catalyst **L1** gave the desired 3-component coupling products in moderate to high yields (Table 1).⁷ After the reaction, **L1** was recovered and a trace amount of **2** were obtained. However, the use of simple cyclopropenes without an acetal moiety gave the desired product in low yield (Scheme 5). Accordingly, we envisioned that we could use cyclopropylzinc intermediates for tandem ring-opening and the allylation reaction, which can be prepared via the carbozincation of cyclopropenes (Scheme 6).⁸ Screening of appropriate catalysts realized the Cu-catalyzed carbozincation of simple cyclopropenes (Scheme 7). However, the use of cyclopropylzinc intermediates generated in situ did not give the desired allylation product. In this context, the allylation reaction might not take place via the carbozincation of cyclopropenes. Furthermore, the use of ketones as electrophiles gave the products in low yield. To overcome these problems, we used a metal catalyst to generate carbene intermediates derived from simple cyclopropenes, which should give non-functionalized allylmetal intermediates. The allylmetal intermediate derived from CPA might decrease the reactivity due to the intramolecular coordination to a metal center (Figure 3). Extensive studies realized the reaction of simple cyclopropenes, organozinc reagents, and ketones in the presence of a Ag-catalyst (Table 2).⁹ The use of non-functionalized cyclo-

**Scheme 5.** Reaction using simple cyclopropene.**Scheme 6.** Proposed mechanism via carbozincation.**Scheme 7.** Cu-catalyzed carbozincation of cyclopropenes.**Table 2.** Ag-catalyzed tandem allylation of aldehydes


Entry	R ¹ , R ² , R ³ , X	Yield (%)
1	Et, Ph, H, O (3a)	91 (4a)
2	Me, Ph, H, O (3a)	64 (4a-Me)
3	<i>i</i> -Pr, Ph, H, O (3a)	59 (4a-i-Pr)
4	Et, 2-MeC ₆ H ₄ , H, O (3b)	68 (4b)
5	Et, 3-MeC ₆ H ₄ , H, O (3c)	83 (4c)
6	Et, 4-MeC ₆ H ₄ , H, O (3d)	95 (4d)
7	Et, 4-MeOC ₆ H ₄ , H, O (3f)	94 (4e)
8	Et, 4-FC ₆ H ₄ , H, O (3g)	83 (4g)
9	Et, 1-naphthyl, H, O (3h)	84 (4h)
10	Et, 2-furyl, H, O (3j)	92 (4j)
11	Et, 2-Phenylethyl, H, O (3k)	71 (4k)
12	Et, 2-Phenylethenyl, H, O (3l)	72 (4l)
13	Et, <i>c</i> -Hex, H, O (3m)	86 (4m)
14	Et, <i>t</i> -Bu, H, O (3n)	46 (4n) ^a
15	Et, Ph, H, NTs (3o)	78 (4o)

a) The product was isolated as a ketone after deprotection of acetal moiety.

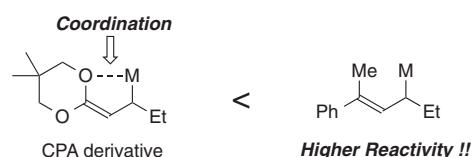
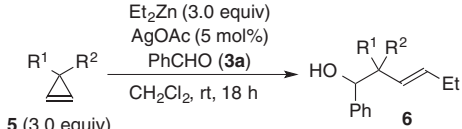
**Figure 3.** Proposed allylmetal intermediates.

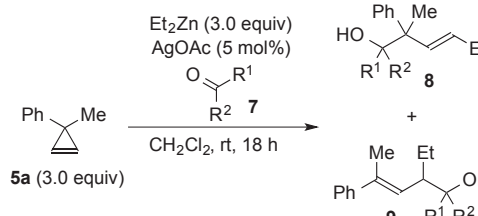
Table 3. Scope of cyclopropenes using Ag-catalyst


Entry	R ¹ , R ²	Yield (%)
1	Ph, Me (5a)	77 (6a) ^a
2	Ph, Et (5b)	63 (6b) ^b
3	Ph, <i>i</i> -Pr (5c)	trace
4	Ph, 1-phenyl-2-methyl-2-propenyl (5d)	32 (6d) ^c
5	Ph, Ph (5e)	56 (6e)
6	4-MeC ₆ H ₄ , Ph (5f)	nd ^d
7	4-FC ₆ H ₄ , Ph (5g)	33 (6g)
8	4-ClC ₆ H ₄ , Ph (5h)	48 (6h)
9	EtO ₂ C, CO ₂ Et (5i)	nd

a) dr = 1.6:1. b) dr = 4:1. c) dr = 5:3. d) Indene product was observed.⁹

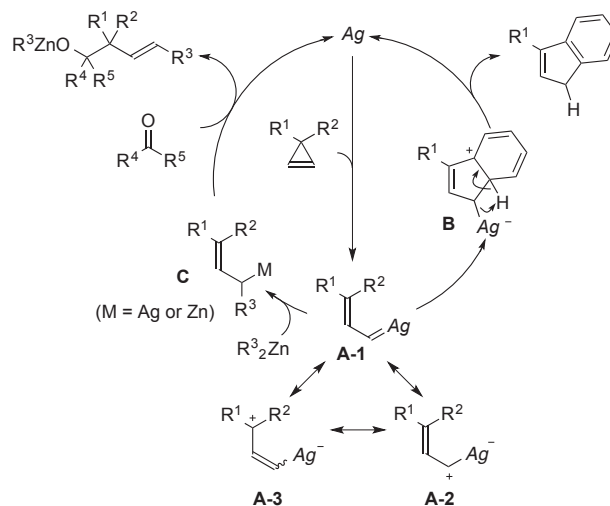
propenes gave the desired products in moderate to high yields (Table 3). Furthermore, ketones participated in the reaction as electrophiles to give α -allylation products as well as γ -allylation products (Table 4). A plausible mechanism shows the reaction of a Ag-carbene intermediate **A-1**, but the generation of an allyl cationic intermediates **A-2** or **A-3** could not be excluded (Scheme 8). The present study showed that the carbenoid intermediates derived from cyclopropenes are electrophilic toward organometallic reagents. We further examined cyclopropanation using cyclopropenes and alkenes via carbene intermediates. During our examination, López and Vicente et al. reported their excellent work on the Zn-catalyzed cyclopropanation of alkenes (Figure 4).¹⁰ Therefore, the use of organozinc reagents may generate carbene intermediates, which would be effective for the tandem allylation reaction as described (Schemes 4, 5, and Table 1).

2.2 Development of Diborylmethane Derivatives. We focused on the vacant orbital on a boron atom, which would be an efficient electron acceptor for the activation of neighboring reactive sites. Accordingly, stable and readily available diborylmethane derivatives should be promising for the present electrophilic activation strategy (Figure 5).¹¹ The first synthesis of diborylmethane derivatives was reported by Matteson et al. DFT calculations for a diborylmethane indicated a large distribution of a LUMO mapping around the diboron moiety, the orbital interaction of which would be suitable for accepting electrons from a nucleophile, such as a base. Alkylboron compounds typically show a lower Lewis acidic character than

Table 4. Ag-catalyzed tandem allylation of ketones


Entry	Ketone 3	Yield (%)
1	Acetone (7a)	67 (8a)
2	Ph-C(=O)-Me (7b)	83 (8b) ^a
3	Ph-C(=O)-CF ₃ (7c)	51 (8c) ^b
4	Cyclopentanone (7d)	86 (8d)
5	Cyclohexanone (7e)	88 (8e)
6	3,4-dihydro-2H-pyran-2-one (7f)	78 (8f)
7	2-Pentanone (7g)	38 (8g), 21 (9g)
8	Ph-C(=O)-Ph (7h)	55 (9h)
9	Bn-C(=O)-Bn (7i)	58 (9i)
10	Bicyclo[1.1.0]butan-2-one (7j)	27 (9j)

a) dr = 2:1. b) dr = 10:9.

**Scheme 8.** Proposed mechanism for Ag-catalyzed allylation using cyclopropenes.

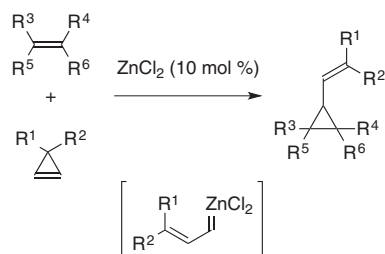


Figure 4. Zn-carbene complex for cyclopropanation of alkenes developed by López and Vicente.

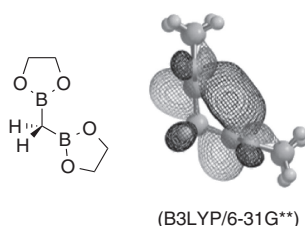
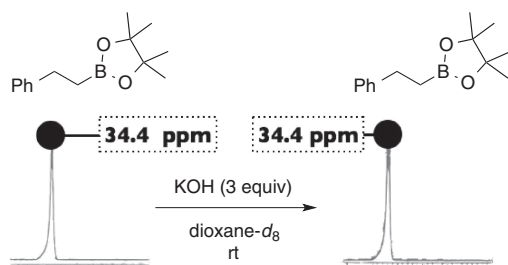
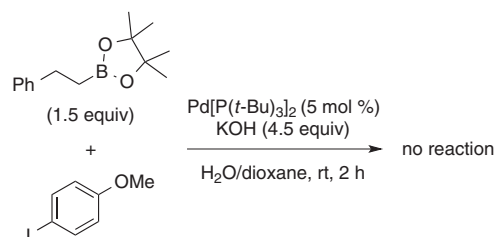


Figure 5. Design of diboryl methane derivative.

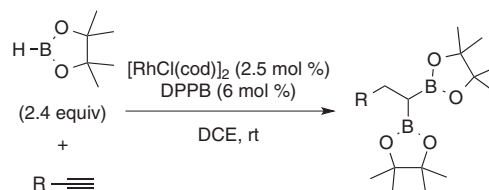


Scheme 9. ^{11}B NMR spectra of monoborylalkane with KOH.

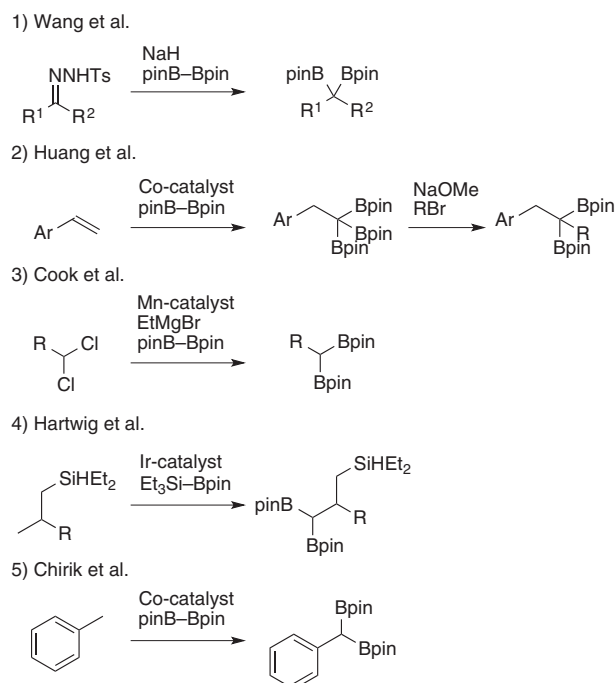
alkenyl or arylboron compounds.¹² Furthermore, stable alkylboronates require a high reaction temperature for the generation of borate intermediates with strong bases. Actually, the reaction of phenethyl pinacol boronate with an excess amount of KOH in dioxane- d_8 did not give borate intermediates at room temperature, which was confirmed by ^{11}B NMR analysis (Scheme 9). The coupling reaction between phenethyl pinacol boronate and 4-iodoanisole did not proceed at room temperature (Scheme 10). This result prompted us to develop 1,1-diborylalkane derivatives; the initial approach to the synthesis of 1,1-diborylalkanes was the Rh-catalyzed sequential hydroboration of 1-alkynes (Scheme 11).¹³ Various examples of the synthesis of 1,1-diborylalkanes have been reported since then (Scheme 12).¹⁴ The reaction of 1,1-diborylalkanes, which are analogues of phenethylpinacolboronate, with an excess amount of KOH in dioxane- d_8 at room temperature showed a new peak in the ^{11}B NMR spectrum at ca. 0 ppm, which indicated the generation of a borate intermediate (Scheme 13). Thus, we considered that 1,1-diborylalkanes should be a favorable candidate for a variety of transformations using organoboron compounds. In this context, we focused on the Suzuki–Miyaura cross-coupling reaction of 1,1-diborylalkanes, which should give a mono-coupling product as well as a di-coupling product. Furthermore, the drawbacks in the Suzuki–Miyaura cross-coupling reaction, i.e., slow transmetalation leading to protoboronation and β -hydride elimination under thermal conditions, would be expected to give various side-products. We



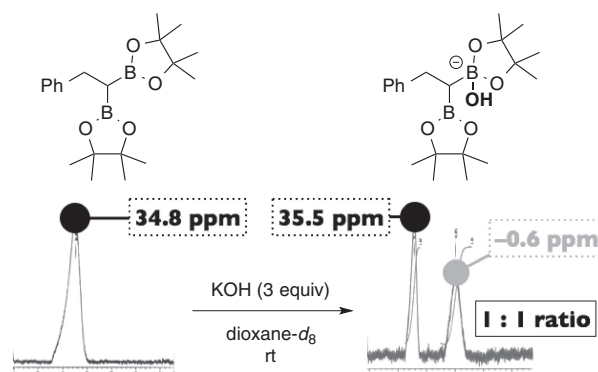
Scheme 10. Coupling reaction of monoborylalkane.



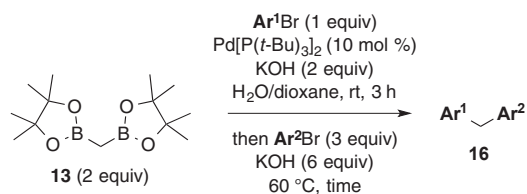
Scheme 11. Rh-catalyzed sequential hydroboration of alkyne.



Scheme 12. Recent examples of diboryl methane derivatives.



Scheme 13. ^{11}B NMR spectra of diborylalkane with KOH.



Scheme 16. One-pot sequential cross-coupling.

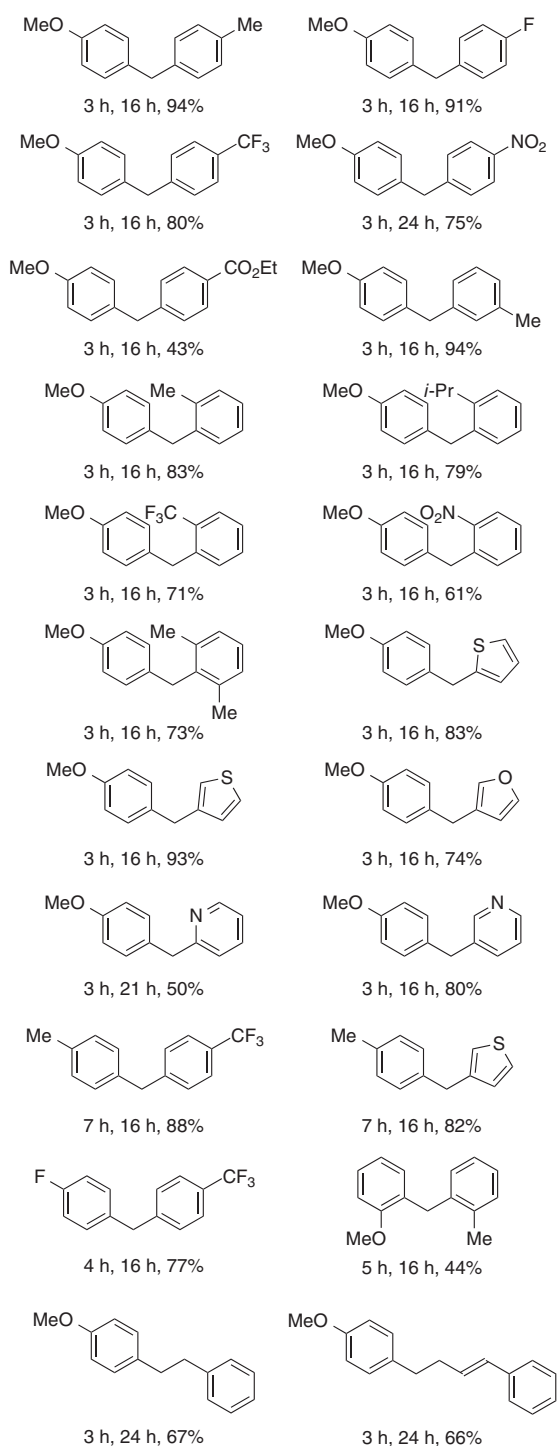
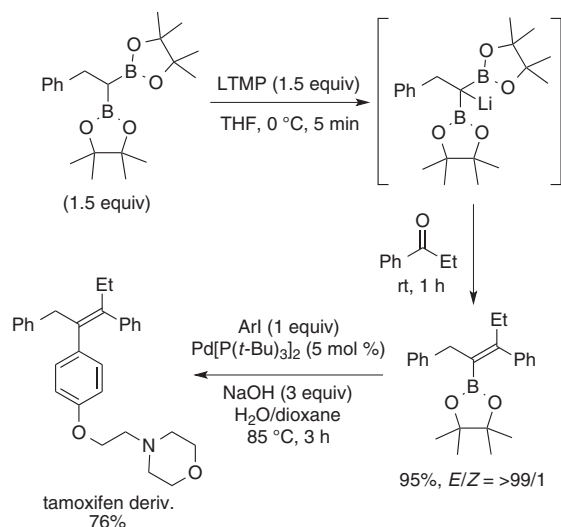
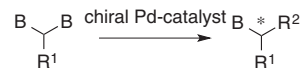


Figure 8. Synthesis of diarylmethane derivatives.

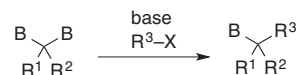


Scheme 17. Stereoselective synthesis of tetrasubstituted alkenes.

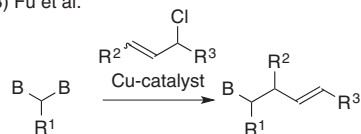
1) Morken et al.



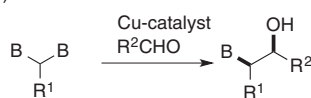
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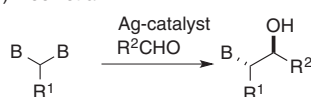
3) Fu et al.



4) Meek et al.



5) Meek et al.



Scheme 18. Diborylmethane derivatives as starting materials.

deprotonation reaction of protic moieties using organometallic reagents and the subsequent coordination of phosphorous moieties to metal centers could generate multinuclear complexes, the major components of which would be oligomers.

We examined the present strategy in the Cu-catalyzed asymmetric conjugate addition reaction.²¹ The reaction of Et_2Zn and chalcone was conducted (Table 5). While the reaction using (*R*)-BINOL as a ligand did not give the desired product (Entry 1), the use of **BP** gave the desired product with high ee (Entries 2 and 3). The protic moieties in a ligand are important; the use of **BP-MOM** dramatically decreased both the yield and ee (Entry 4). The optimized conditions achieved excellent yield

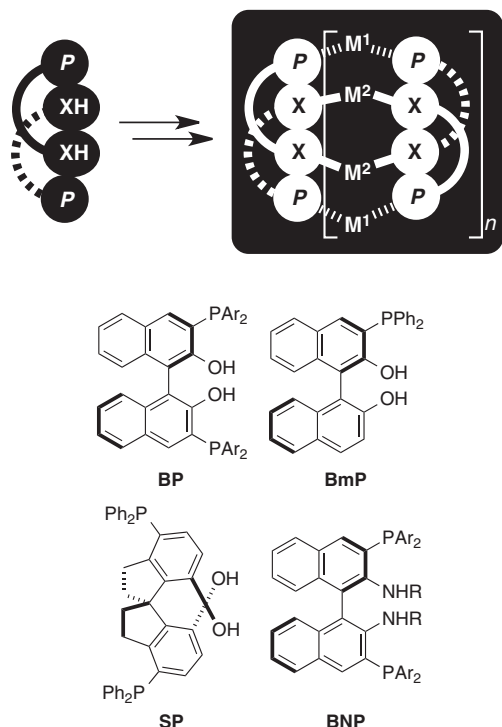


Figure 9. Our strategy and representative ligands.

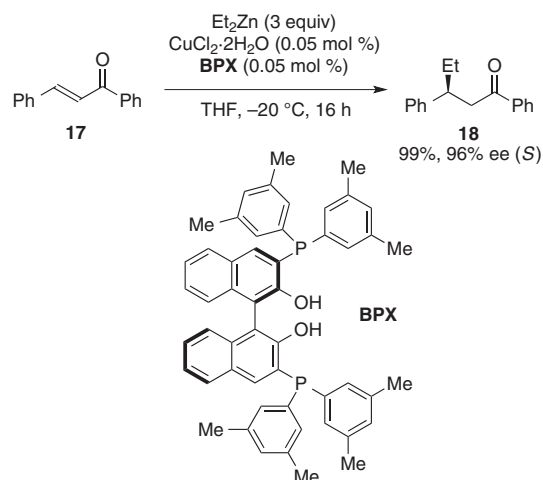
with high ee (Entries 5 and 6). The use of mono-phosphorous ligand **BmP** decreased the yield and ee (Entry 8). Finally, the use of (*R*)-BINOL and PPh_3 did not show enantioselectivity (Entry 9). The best ligand was **BPX**. The reaction of chalcone and organozinc reagents in the presence of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (0.05 mol %) and **BPX** (0.05 mol %) achieved excellent catalytic performance; the desired product was obtained in 99% yield with 96% ee (Scheme 19, Figure 10).²² In contrast, the use of SPINOL derivative **SP** as a ligand gave the opposite major enantiomers selectively (Scheme 20).²³ Ligands bearing phosphorous moieties at the *ortho*-position to a hydroxyl group gave the (*S*)-product as a major enantiomer and ligands bearing phosphorous moieties at the *para*-position to a hydroxyl group gave the (*R*)-product as a major enantiomer. The BINAM-based ligands **BNmP** unexpectedly gave the desired products as the opposite major enantiomers compared to those using BINOL-based ligands (Scheme 21).²⁴

The ESI-MS analysis of Cu/Zn-complexes derived from **BP** suggested the generation of tetrametallic complexes as a major proportion of oligomeric complexes (Figure 11). The deprotonation reaction using organoaluminum reagents also showed a dinuclear complex by ESI-MS analysis (Figure 12). The present Cu/Al-complex can catalyze the asymmetric conjugate addition reaction of organoaluminum reagents to enones and acrylamide derivatives (Scheme 22, Figure 13).²⁵ Acrylamide derivatives are synthetic intermediates for the synthesis of natural products (Scheme 23). Furthermore, the first example of the asymmetric conjugate addition to acyclic tri-substituted enones successfully created a chiral quaternary stereogenic center (Scheme 24, Figure 14).²⁶ The reaction using trisubstituted unsaturated ketoesters gave chiral furanones bearing a quaternary carbon (Scheme 25, Figure 15).²⁷ The use of Et_3Al

Table 5. Multinuclear Cu/Zn-catalyzed conjugate addition

Entry ^{a,b}	L (y mol %)	Time (h)	Yield, ee (%) ^c
1	(<i>R</i>)-BINOL (5)	24	—
2	BP (5)	1	67, 92
3	BP (10) ^d	48	23, 68
4	BP-MOM (5)	72	12, 10
5	BP (5)	0.5	96, 91
6	BP (5)	0.5	>98, 94
7	BmP (5)	24	21, 65
8	BmP (10) ^d	24	30, 71
9	(<i>R</i>)-BINOL (5) ^e	16	89, <1

a) Entries 1–5, 0 °C; Entries 6–9, –20 °C. b) Entries 1–4, $x = 1.5$; Entries 5–9, $x = 3.0$. c) NMR yields. The ee was determined by chiral HPLC analysis. d) $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (5 mol %) was used. e) PPh_3 (10 mol %) was used.



Scheme 19. Cu-catalyzed asymmetric conjugate addition of organozinc reagents to chalcone.

or AeMe_2Al gave the desired Ar-adduct in good yield with moderate to excellent enantioselectivity (Figure 16). While the mechanism of the present catalysts is unclear, reactive intermediates bearing a Lewis acidic moiety might increase the electrophilicity of the active center.

The present strategy can be applied to the Pd-catalyzed reaction using organozinc reagents. ESI-TOF-MS analysis of Pd/Zn-complexes suggested the generation of a hexa-metallic complex (Figure 17). After extensive studies, the Pd-catalyzed asymmetric ring-opening reaction of oxabicyclic alkenes **25a–25i** took place to give the desired products **26a–26i** in high

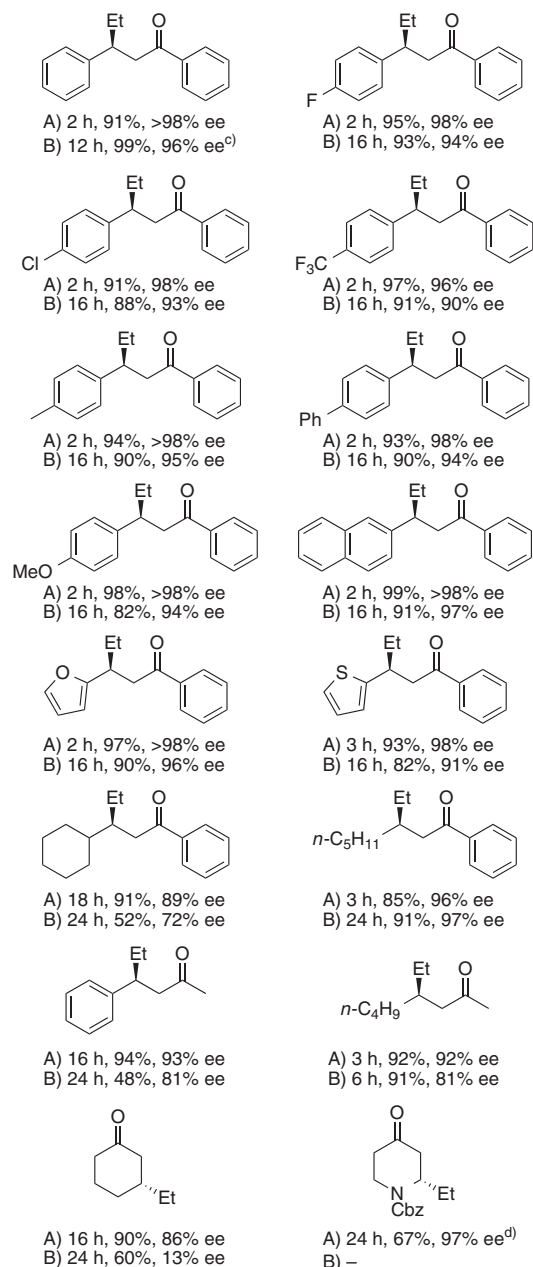
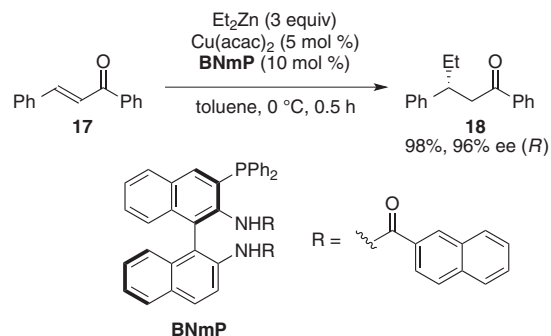
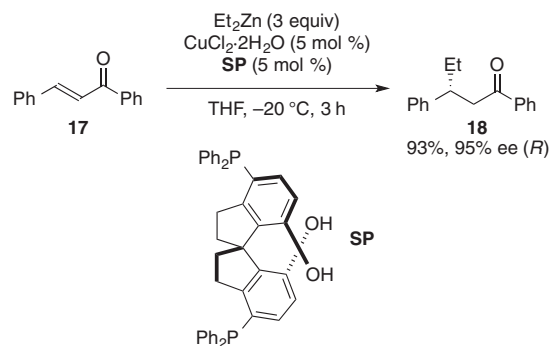


Figure 10. Cu-catalyzed enantioselective conjugate addition of organozinc reagents^{a,b)}. a) Enone (0.5 mmol) was used; Condition A: CuCl₂·2H₂O (0.5 mol %), **BPX** (0.5 mol %), Condition B: CuCl₂·2H₂O (0.05 mol %), **BPX** (0.05 mol %). b) Isolated yields. NMR yields were described in parentheses. The ee was determined by the chiral HPLC analysis or chiral GC analyses. The absolute configuration was determined by comparing the reported specific rotation data or retention time of HPLC analyses. c) (*E*)-Chalcone (5 mmol) was used. d) CuBr·SMe₂ (12 mol %) and **BPX** (12 mol %) in toluene.

yields with excellent enantioselectivities (Table 6).²⁸ The present reaction proceeds via the enantioselective carbopalladation of an olefinic moiety followed by β-elimination for ring-opening of the oxabicyclic framework.²⁹



Scheme 21. Effect of BINAM derivative **BNmP**.

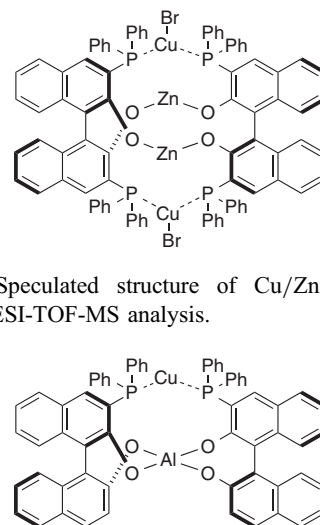


Figure 11. Speculated structure of Cu/Zn-complex detected by ESI-TOF-MS analysis.

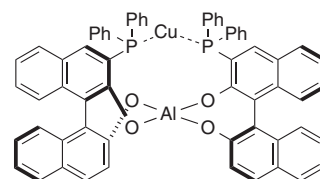
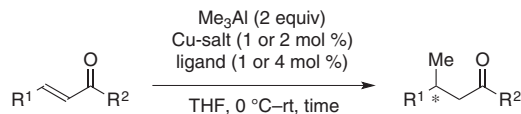


Figure 12. Speculated structure of Cu/Al-complex detected by ESI-TOF-MS analysis.



Scheme 22. Me₃Al for tertiary stereogenic center.

4. Conclusion

In conclusion, the incorporation of Lewis acidic moieties in reactive intermediates realized efficient transformation. The electrophilic carbenoid intermediates can promote the 3-component coupling reaction via the tandem nucleophilic

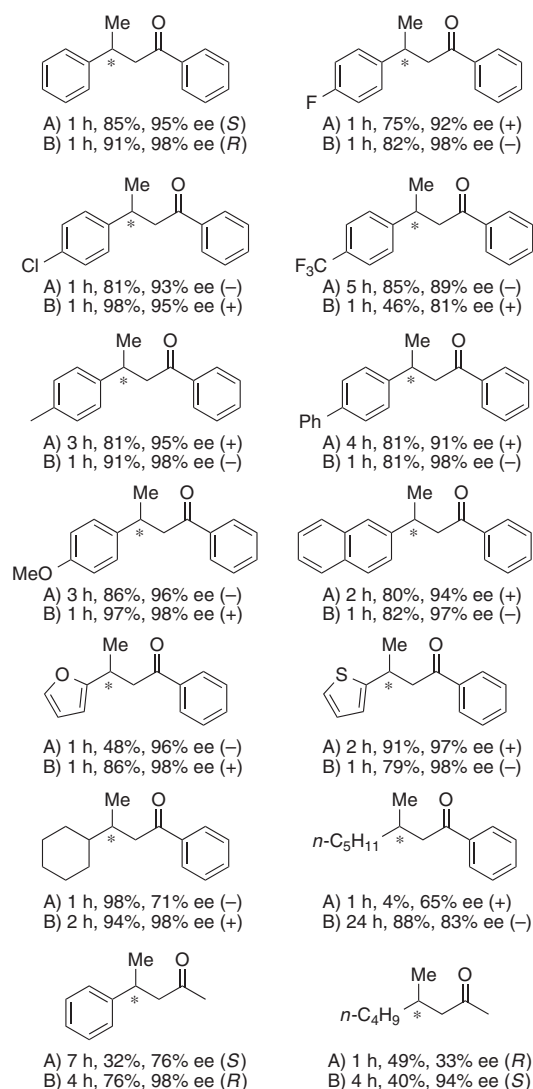
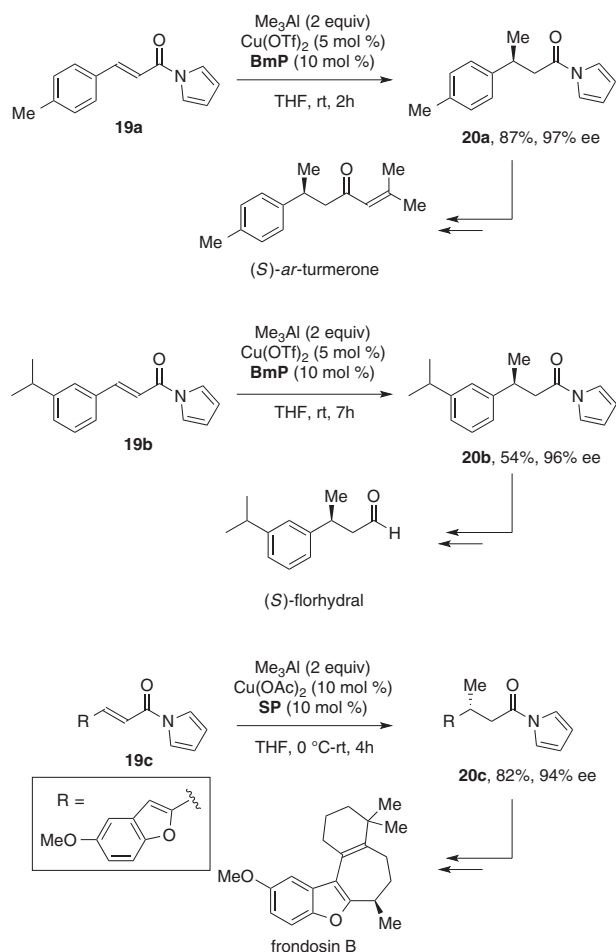


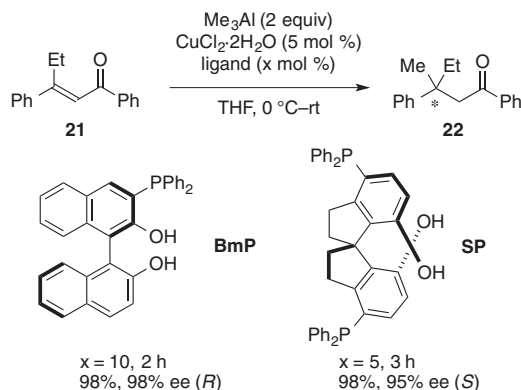
Figure 13. Addition of Me_3Al to various enones^{a,b}. a) Isolated yields. Ee was determined by chiral HPLC or GC analysis. b) Conditions A: $\text{Cu}(\text{OTf})_2$ (2 mol %) and **BmP** (4 mol %); Conditions B: $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (1 mol %) and **SP** (1 mol %).

addition reaction to carbenoid intermediates and electrophiles in the presence of a catalyst. The neighboring boron moiety showed the unique reactivity of diborylmethane derivatives to give borate intermediates at room temperature, which is suitable for the Suzuki–Miyaura cross-coupling reaction. The multi-metallic complex based on BINOL, SPINOL, and BINAM can catalyze the asymmetric addition reaction of organometallic reagents. The Cu/Al-complex realized the creation of a chiral quaternary carbon via the conjugate addition reactions. The present powerful catalysis should contribute to efficient transformations that cannot be established with conventional catalyst systems.

The author is grateful to Profs. Takanori Shibata, Kuniaki Tatsuta, and Yutaka Ukaji for their generous support and fruitful comments. I also appreciate Ms. Mika Ogawa, Mr. Munenao Hirokami, Ms. Ai Kawachi, Mr. Yusuke Kuwata, Mr. Keisuke



Scheme 23. Synthetic intermediates via conjugate addition.



Scheme 24. Enantioselective addition of Me_3Al to trisubstituted enones.

Tanaka, Mr. Kazunari Takeuchi, Mr. Takahiro Ohkubo, Mr. Daisuke Hamada, Ms. Sayuri Yakeishi, Mr. Akira Sakamoto, Mr. Takafumi Ishioka, Mr. Ryotaro Takayama, Dr. Takeo Nakano, Mr. Fumiya Kurosawa, Mr. Koichi Kawamura, Ms. Saori Tani, Mr. Tomoki Kobayashi, Mr. Takuya Sekiya, Mr. Hiroyuki Hayashi, and Ms. Saori Shitaya. This work was supported by a JST PRESTO program, Grant-in-Aid for Young Scientists from the Japan Society for the Promotion of Science, The Kurata Memorial Hitachi Science and Technology Foundation, and Hokuriku Bank.

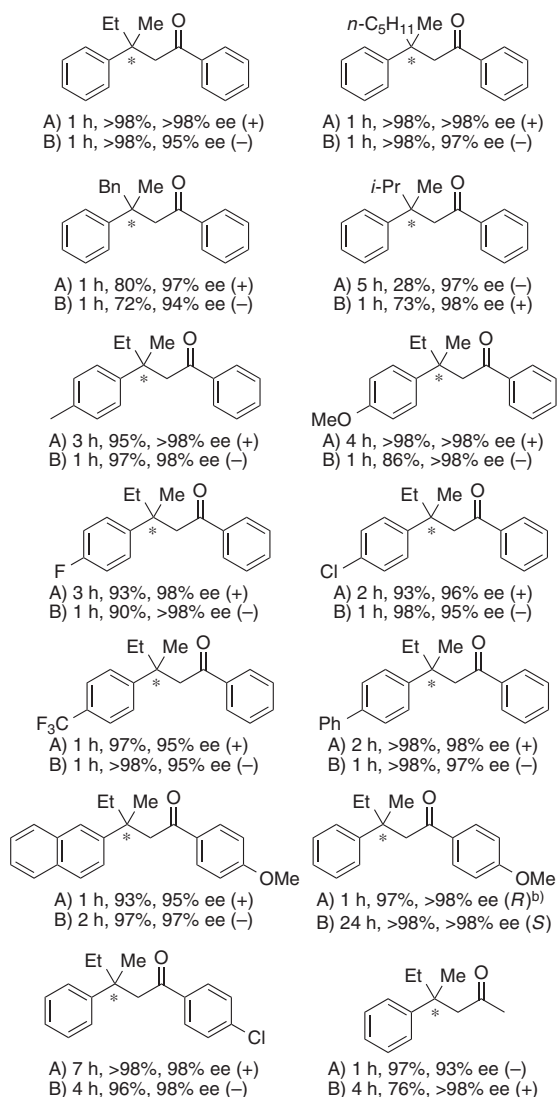
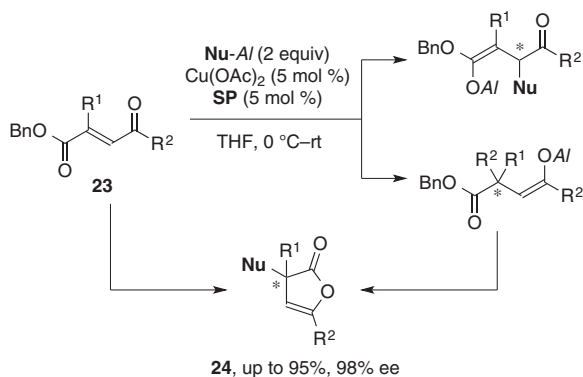


Figure 14. Creation of chiral quaternary stereogenic center^{a)}. a) Isolated yields. Ee was determined by chiral HPLC analysis. b) The absolute configuration was determined after the conversion to the known carboxylic acid.



Scheme 25. Cu-catalyzed enantioselective conjugate addition of organoaluminum reagents to unsaturated ketoesters.

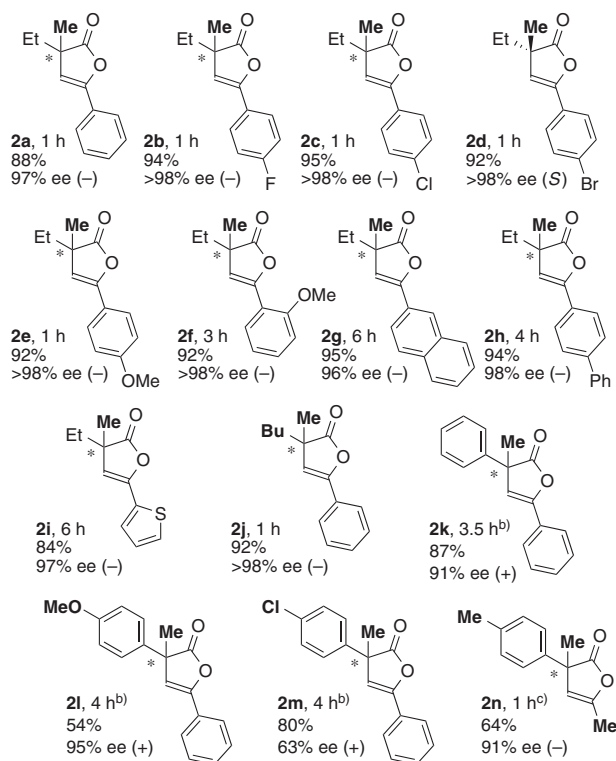


Figure 15. Enantioselective synthesis of furanones^{a)}. a) $\text{Cu}(\text{OAc})_2$ (5 mol %), **SP** (5 mol %), Me_3Al (1.5 equiv) were used. b) **BmP** (10 mol %) and Me_3Al (2 equiv) were used. c) The corresponding ethyl ester, $\text{Cu}(\text{OAc})_2$ (5 mol %), **BmP** (10 mol %), Me_3Al (6 equiv), and ether as a solvent were used.

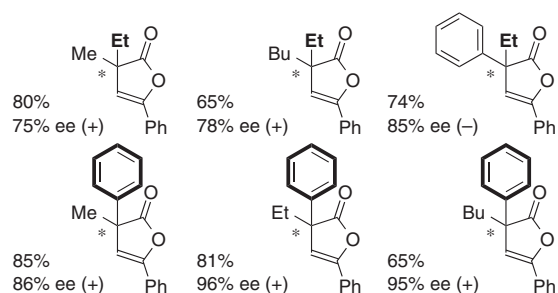


Figure 16. Use of Et_3Al and PhMe_2Al for furanones.

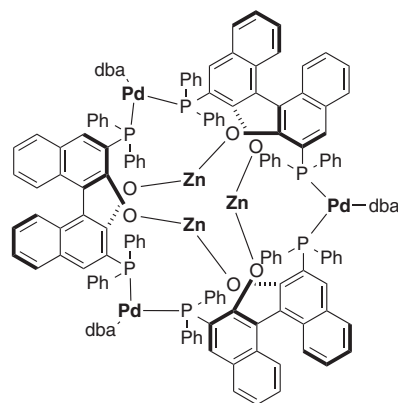


Figure 17. Speculated structure of Pd/Zn-complex (dba = dibenzylideneacetone).

Table 6. Multinuclear Pd/Zn-catalyzed asymmetric ring-opening of oxabicyclic alkenes

Entry ^{a)}	Time (h)	Product	Yield, ee (%) ^{b)}
		 26a-i	
1	16	26a , R = H	87, 97 ^{d)}
2	15	26b , R = Me	94, 99 ^{d)}
3	18	26c , R = OMe	95, 98 ^{e)}
4	48	26d , R = F	44, 81 ^{d)}
		 26e-i	
5	5	26e , R = Me	96, 98 ^{d)}
6	24	26f , R = OMe	85, 99 ^{d)}
7	4	26g , R = -OCH ₂ O-	98, 97 ^{d)}
8	22	26h , R = F	91, 98 ^{e)}
9	48	26i ^{c)}	83, 97 ^{e)}
		 26i ^{c)}	

a) Me₂Zn (1.5 equiv in Entry 1; 3 equiv in Entries 2, 3, 5, 6, and 7; 5 equiv in Entries 8 and 9) was used. b) Isolated yields. The enantiomeric excess was determined by chiral HPLC analyses. c) Absolute configuration was not determined. d) A hexane solution of 1 M Me₂Zn (3 equiv) was used. e) A toluene solution of 1 M Me₂Zn (3 equiv) was used.

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